

PAIRSELECT: TRAINING-FREE TOKEN SELECTION FOR EFFICIENT WEAKLY SUPERVISED VIDEO ANOMALY DETECTION

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ABSTRACT

Detecting rare events requires sustained observation, but not full computation for every internal token. We introduce PairSelect, a training-free operator that retains local, magnitude-guided representatives after six dense transformer blocks, without mixing their vectors or reducing input coverage. To assess generalization, we evaluate four weakly supervised video anomaly detection systems: VideoMAEv2-B, VideoMAE-B, and TimeSformer-B paired with a common UR-DMU detection-head architecture, and CLIP ViT-B/16 paired with DSANet, a recent state-of-the-art detector. Each UR-DMU head is trained separately on its encoder’s full dense-feature training set, and all four systems reuse their detector weights across token budgets. Across UCF-Crime and XD-Violence, 40% nominal token deletion preserves near-dense performance in most configurations, with batch-32 encoder speedups of 1.22–1.28 $\times$. With only mild performance degradation in most configurations, increasing deletion to 60% delivers up to 1.47 $\times$ encoder acceleration. These results demonstrate that PairSelect generalizes across visual representations and detection architectures, improving encoder efficiency without compression-specific retraining.

Index Terms— video anomaly detection, weak supervision, token selection, training-free compression, efficient inference

1. INTRODUCTION

Video anomaly detection faces an asymmetric computational problem: unusual events are scarce, yet the system must process long stretches of ordinary activity to find them. Weakly supervised video anomaly detection (WSVAD) learns from video-level labels without knowing which snippets contain anomalies [1, 2]. Contemporary systems combine pretrained visual representations with temporal detectors, from dual-memory models [3] to CLIP-based semantic alignment [4, 5]. Their representations differ, but they share a deployment question: can an existing detector use less encoding computation without another training cycle?

Reducing the number of sampled clips also changes the temporal observations available to the detector. We instead preserve the original sampling schedule and reduce computation inside the visual encoder. *The rarity of anomalous events does not imply that every internal token carries unique information.* Video redundancy supports masked pretraining [6, 7], while token sampling and merging demonstrate opportunities to process fewer representations [8, 9]. This suggests allowing every token to participate in early contextualization, then continuing with locally selected representatives.

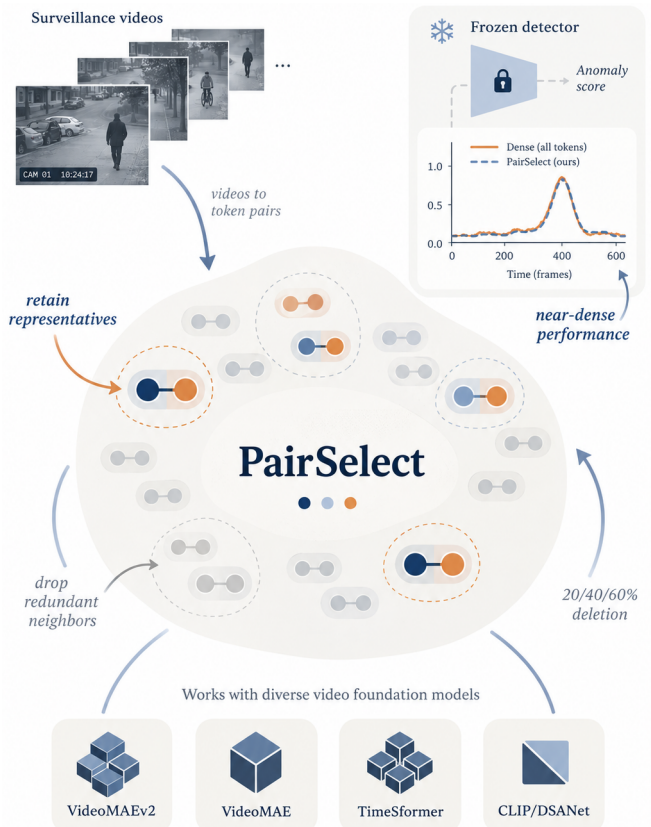


Fig. 1. Conceptual overview of PairSelect: local representative selection at three deletion budgets with frozen detection across four systems. Score curves are schematic, not measured results.

We propose PairSelect, a non-mixing token selection operator for frozen WSVAD systems (Figs. 1 and 2). Local pairing defines where candidates compete, group quotas distribute the budget, and activation magnitude determines member priority. Selected vectors are gathered without averaging. The same principle applies to video-patch representations and CLIP image patches; divided space-time attention instead selects complete spatial trajectories. Neither the visual encoder nor the detector is retrained after insertion or when the budget changes.

Our evaluation jointly covers four systems: VideoMAEv2-B,

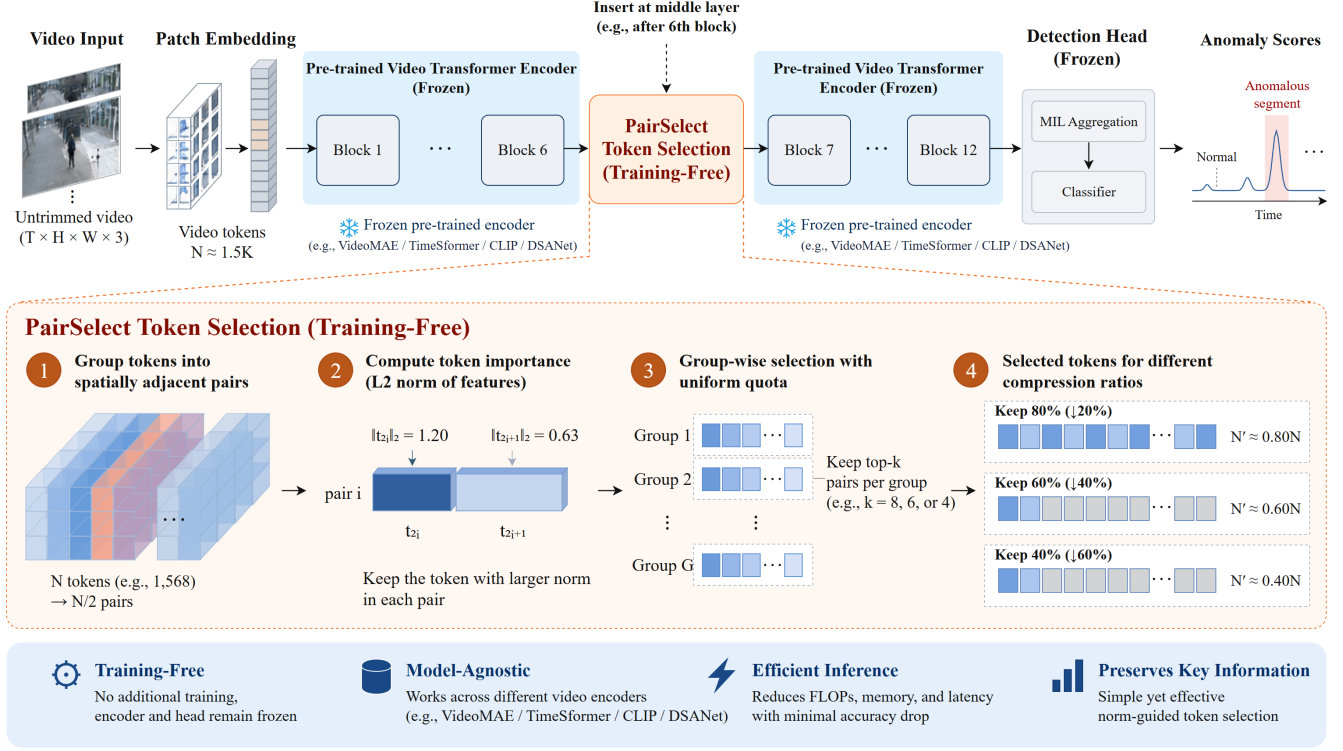


Fig. 2. Overview of PairSelect. Six dense blocks contextualize all tokens before local, magnitude-guided selection reduces the input to blocks 7–12. Group quotas control pair-member selection; divided attention retains complete spatial trajectories. The four systems reuse their original temporal sampling and detector weights; CLIP applies pair selection per image and preserves CLS.

VideoMAE-B, and TimeSformer-B with their respective UR-DMU heads, and CLIP ViT-B/16 with the published DSANet weights. DSANet is a recent high-performance, state-of-the-art-level WS-VAD system [5], making this a test of transfer across both visual representations and detector designs. We contribute the local selection rule, its architecture-aware application, and a complete three-budget quality–efficiency study on two datasets. At 40% nominal deletion, most configurations retain near-dense quality; 60% deletion offers greater encoder throughput with generally mild quality costs.

2. RELATED WORK

Weakly supervised anomaly detection. Multiple-instance ranking [1], temporal feature magnitudes [2], and uncertainty-regulated memory [3] address weak supervision. VadCLIP adapts vision–language representations [4]; DSANet disentangles normality and abnormality through semantic alignment [5]. Our four-system study uses both UR-DMU and DSANet without changing their detector weights for compression. It is visual-only on both UCF-Crime and the audiovisual XD-Violence benchmark [10].

Efficient visual representations. VideoMAE-family encoders use masked video pretraining [6, 7], whereas TimeSformer separates spatial and temporal attention [11]. DynamicViT learns token predictors [12], EViT reorganizes token representations [13], and EVAD introduces task-aware token dropout for action detection [14]. ATS and ToMe support reduction without additional training [8, 9]. PairSelect focuses on locally constrained, non-mixing selection that can be inserted into an existing anomaly detector. In contrast,

ESOM and LAVIDA reduce tokens within multimodal language-model pipelines [15, 16]. Our target is the visual-encoder cost of frozen temporal detection systems.

3. PAIRSELECT

3.1. Direct insertion into frozen systems

Let $E_{1:b}$ and $E_{b+1:L}$ be the prefix and suffix of a pretrained visual encoder, R its readout, and H its temporal detection head. For scheduled visual inputs $\{v_t\}_{t=1}^M$,

$$z_t^{(r)} = R(E_{b+1:L}(S_r(E_{1:b}(v_t)))) , \quad \hat{y}^{(r)} = H(\{z_t^{(r)}\}_{t=1}^M). \quad (1)$$

Only the selection operator S_r is inserted, with retention ratio r . All four visual encoders use $L = 12$, $b = 6$. The original input schedule, feature dimension, temporal aggregation, and head weights are preserved. For the three video backbones, each UR-DMU head is trained once on the corresponding dense training features; DSANet

Table 1. Actual token counts entering blocks 7–12. Headers specify nominal deletion; CLS is included when present. Video encoders count tokens per clip; CLIP counts per image.

Visual encoder + detector	Dense	20%	40%	60%
VideoMAEv2-B + UR-DMU	1568	1254	941	627
VideoMAE-B + UR-DMU	1568	1254	941	627
TimeSformer-B + UR-DMU	1569	1233	897	561
CLIP ViT-B/16 + DSANet	197	158	119	79

Table 2. Four-system quality at every budget (0–100). Each cell reports the indicated two frame-level metrics; bold identifies the primary metric, not a ranking. On XD, the video systems use trapezoidal PR-AUC and DSANet uses step AP. CI is the paired 95% interval for the primary-metric change at 40% deletion, in percentage points.

System	Dense	Delete 20%	Delete 40%	Delete 60%	CI of Δ_{40}
<i>UCF-Crime: ROC-AUC / step AP (primary: ROC-AUC)</i>					
VideoMAEv2–UR-DMU	80.63 / 19.29	80.90 / 19.33	81.02 / 19.60	80.95 / 19.77	[-0.94, +1.90]
VideoMAE–UR-DMU	80.55 / 18.82	80.54 / 18.83	80.39 / 18.99	79.50 / 18.77	[-0.75, +0.50]
TimeSformer–UR-DMU	78.68 / 20.37	78.74 / 20.71	78.68 / 20.90	79.24 / 21.90	[-1.12, +1.11]
CLIP–DSANet	88.87 / 34.74	88.54 / 34.10	87.97 / 34.36	87.06 / 34.77	[-2.07, +0.21]
<i>XD-Violence: trapezoidal PR-AUC / step AP (primary: bold)</i>					
VideoMAEv2–UR-DMU	76.18 / 70.33	76.52 / 70.97	76.51 / 71.30	75.81 / 71.11	[-0.06, +0.75]
VideoMAE–UR-DMU	70.34 / 64.86	70.66 / 65.11	70.94 / 64.51	71.11 / 63.47	[+0.02, +1.17]
TimeSformer–UR-DMU	76.69 / 70.86	77.21 / 71.63	77.37 / 72.04	77.05 / 72.03	[+0.22, +1.13]
CLIP–DSANet	85.60 / 85.60	85.49 / 85.49	85.35 / 85.35	84.38 / 84.38	[-1.00, +0.50]

uses the authors’ released detector weights. *Training-free* refers to inserting and changing the compression operator, not to training the original systems.

Local redundancy motivates comparisons among neighboring patches, while selection avoids introducing additional vector mixtures at the insertion point. Activation magnitude provides an inexpensive preference, inspired by magnitude-aware sampling [8]. It is not an anomaly probability [17]; the local quota prevents an unconstrained global ranking from concentrating the entire budget in a few regions.

3.2. Local quotas and member priority

For VideoMAE, VideoMAEv2, and CLIP, spatially adjacent horizontal patches form pairs ordered by native coordinates. Each consecutive group contains at most ten patch tokens (five pairs). A group of n_g tokens receives

$$q_g(r) = \lfloor rn_g + \frac{1}{2} \rfloor, \quad r \in \{0.8, 0.6, 0.4\}. \quad (2)$$

For pair j with members $(x_{j,0}, x_{j,1})$, define

$$a_j = \arg \max_{u \in \{0,1\}} \|x_{j,u}\|_2, \quad s_j = \max_{u \in \{0,1\}} \|x_{j,u}\|_2. \quad (3)$$

Let $w_j = x_{j,a_j}$ and $\ell_j = x_{j,1-a_j}$. If π sorts the group’s m_g pairs by descending s_j , the priority sequence is

$$\mathcal{P}_g = (w_{\pi_1}, \dots, w_{\pi_{m_g}}, \ell_{\pi_1}, \dots, \ell_{\pi_{m_g}}). \quad (4)$$

The first q_g entries are retained and restored to native index order before gathering. Ties preserve native order. At 20% and 40% deletion, every pair in a full group retains a representative, with three or one additional members; 60% deletion retains four of five representatives. Selected vectors are neither averaged nor rescaled. CLIP applies the rule independently per image and always retains CLS, yielding the counts in Table 1.

3.3. Structure-aware adaptation

For TimeSformer, the selection unit is a complete spatial trajectory of T tokens, scored by its maximum member norm. Each group of ten trajectories retains its top q_g . The current bridge rounds the total spatial count down to a multiple of native width W , removing excess quota from tail groups:

$$P' = W \left\lfloor \frac{\sum_g q_g(r)}{W} \right\rfloor, \quad K = TP' + 1. \quad (5)$$

With $P = 196$, $T = 8$, and $W = 14$, the three budgets retain 154, 112, and 70 trajectories plus CLS. Actual token deletion is 21.41%, 42.83%, and 64.24%; this alignment constraint is implementation-specific. Video-encoder bridges use a fixed quota-slot skeleton with approximate spatial anchors while retaining trajectory time indices. CLIP gathers the already position-embedded patch states in native order and preserves its CLS readout. These adaptations keep the detector interface unchanged.

4. EXPERIMENTS

4.1. Four-system evaluation

We evaluate on UCF-Crime [1] (1,610 training and 290 test videos) and XD-Violence [10] (3,950 accepted training and all 800 test videos; four declared training files fail source-integrity checks). VideoMAEv2-B, VideoMAE-B, and TimeSformer-B use pretrained encoders and separate UR-DMU heads trained on the corresponding complete accepted training sets. The heads use 200 temporal bins, 64 normal and 64 anomalous bags per step, Adam with learning rate 10^{-4} and weight decay 5×10^{-5} , and the final checkpoint after 3,000 steps. CLIP ViT-B/16 uses the authors’ released UCF/XD DSANet checkpoints [5], with no DSANet training in this study. All four systems reuse their dense detector at every budget.

Each system preserves its native input schedule and temporal scoring protocol. VideoMAEv2-B and VideoMAE-B use 16-frame inputs, TimeSformer-B uses eight-frame inputs, and CLIP ViT-B/16 processes sampled frames under its fixed raw-video protocol, all at 224×224 resolution. Detector weights and scoring rules remain fixed across token budgets, with the coarse branch used for DSANet. The CLIP dense reference is extracted from the same raw videos as its compressed counterparts, ensuring that the comparison uses the evaluated raw-video pipeline rather than a score obtained from externally released features.

UCF uses frame ROC-AUC as primary metric and also reports step AP. On XD, the three video systems use trapezoidal PR-AUC with the fixed 16-frame prefix alignment, while DSANet uses step AP. Table 2 separates both integration definitions. Paired video-level, weak-label-stratified bootstrap uses 10,000 resamples; all-budget intervals are retained with the result exports.

4.2. Quality across systems and budgets

At 40% nominal deletion, PairSelect is nearly lossless in most configurations. The three video systems change UCF AUC by +0.39,

Table 3. A100 encoder efficiency using XD training inputs. Block GFLOPs cover principal blocks per input. B32 measures prepared GPU input to encoder output, including selection (clips/s for video encoders; images/s for CLIP). Speedups use matching dense baselines. Bold marks the optimum within each system (arrows give direction).

System	Delete	Block GFLOPs↓	B32 ms↓	Inputs/s↑	Encoder × ↑
VideoMAEv2–UR-DMU	Dense	356.99	1166.66	27.43	1.000
	20%	313.99	1065.11	30.04	1.095
	40%	274.74	952.31	33.60	1.225
	60%	239.00	876.99	36.49	1.330
VideoMAE–UR-DMU	Dense	356.99	1160.05	27.58	1.000
	20%	313.99	1066.04	30.02	1.088
	40%	274.74	930.12	34.40	1.247
	60%	239.00	841.54	38.03	1.378
TimeSformer–UR-DMU	Dense	389.81	929.69	34.42	1.000
	20%	347.15	837.03	38.23	1.111
	40%	305.01	727.84	43.97	1.277
	60%	263.40	632.58	50.59	1.470
CLIP–DSANet	Dense	34.89	81.75	391.42	1.000
	20%	31.33	74.66	428.59	1.093
	40%	27.82	66.95	477.95	1.221
	60%	24.27	59.10	541.45	1.383

−0.16, and approximately 0.00 points, and XD PR-AUC by +0.32, +0.61, and +0.68. DSANet changes UCF AUC by −0.90 and XD step AP by −0.24 points, retaining 87.97 AUC and 85.35 AP. UCF step AP is also stable: 19.29→19.60, 18.82→18.99, 20.37→20.90, and 34.74→34.36 across the four systems. No compression-specific training is used to obtain these results.

Increasing deletion to 60% generally incurs mild degradation while preserving high detection quality. Several configurations improve, and the three largest primary-metric drops are DSANet/UCF (−1.80), DSANet/XD (−1.22), and VideoMAE/UCF (−1.05 points). This extends the usable reduction range across video-pretrained and image-text-pretrained representations and across UR-DMU and DSANet. The two XD metrics need not move together: VideoMAE at 60% gains 0.78 PR-AUC points but loses 1.39 step-AP points; both are retained in the table. Near-lossless describes point estimates rather than a blanket statistical guarantee: among DSANet budgets, only XD at 20% has its 95% lower bound above the predeclared −0.5-point margin.

4.3. Encoder efficiency

Table 3 compares batch-32 encoder efficiency across budgets using XD training inputs. Encoder timing includes embedding, selection, gathering, all blocks, and readout. Video-encoder B32 tests use two fixed training videos, 20 warmups and 40 synchronized samples; CLIP uses three processes, 20 warmups and 40 timed samples per setting. Arithmetic values cover the principal twelve blocks (one MAC is two FLOPs), not a full-model profiler estimate.

We evaluate encoder efficiency using speedup relative to each system’s dense counterpart at the same batch size. At batch size 32, PairSelect achieves encoder speedups of 1.22–1.28× at 40% nominal token deletion while preserving near-dense detection performance in most configurations. Increasing deletion to 60% raises

encoder speedups to 1.33–1.47×, with generally mild performance degradation. These results highlight a favorable accuracy–efficiency trade-off: PairSelect reduces internal token computation while preserving input coverage and reusing the same detector weights across budgets.

5. DISCUSSION AND FUTURE WORK

The four-system results show that PairSelect enables frozen detectors to identify anomalous frames with fewer internal tokens and no per-budget retraining. The near-dense performance at 40% deletion in most configurations (Table 2) provides empirical evidence that the compressed representations retain anomaly-discriminative information. Dense contextualization followed by local, magnitude-guided selection reduces internal computation while preserving the observation schedule. We report the complete fixed budget ladder to characterize this detection–efficiency trade-off across systems. These findings motivate future compression-aware WSVAD training and joint optimization of representations and selection.

6. CONCLUSION

PairSelect enables efficient frame-level anomaly detection by retaining contextualized token representatives within pretrained visual encoders. Across four encoder–detector systems on UCF-Crime and XD-Violence, the results support preservation of anomaly-discriminative information without compression-specific retraining. At 40% nominal deletion, most configurations retain near-dense detection performance; at 60%, encoder acceleration reaches 1.47× with generally mild performance degradation. Together, these findings demonstrate a transferable route to lower-cost visual encoding for anomaly detection while preserving input coverage.

7. COMPLIANCE WITH ETHICAL STANDARDS

This research uses existing UCF-Crime and XD-Violence benchmarks and does not introduce participant recruitment or newly collected personal data. The experiments concern offline benchmark evaluation, not deployment decisions about individuals.

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